

UCT41042 – Practical for Introduction to Smart System
Department of Information & Communication Technology
Faculty of Technology
South Eastern University of Sri Lanka
Lab Sheet No: 08

Date: 25.05.2026

Title: Smart Garbage Bin Weight Monitoring System using ESP32, HX711 & Blynk.

Aims:

To design and implement a smart garbage bin monitoring system using an ESP32 microcontroller and HX711 load cell amplifier, where the bin weight is continuously measured and displayed on an I2C LCD and the Blynk IoT platform. The system provides visual alerts using an RGB LED to indicate garbage levels (low, medium, full) and detects overweight conditions, thereby optimizing waste collection efficiency.

Task 01: Smart Home Automation System Setup

1. Project Setup in Wokwi / VS Code
 - a. Name the project **SmartGarbage_ESP32_Blynk**.
2. Circuit Design
 - a. Sensor Connection
 - i. Connect the HX711 Load Cell Amplifier to the ESP32.
 - b. Output Devices Connection
 - i. Connect RGB LED and LCD(I2C) display to the ESP32.
3. Library Integration
4. Blynk Configuration
 - a. Create a New Template in Blynk (IoT platform)
 - b. Assign virtual pins
 - c. Create a New Device
 - d. Add widgets to display values.
 - e. Obtain Template ID, Template Name and Auth Token

5. Programming Logic Development in Wokwi / VS Code

- a. Initialize Wi-Fi and Blynk connection
- b. Initialize HX711 and LCD
- c. Read weight from load cell using HX711
- d. Show weight on LCD
- e. RGB LED Logic (Level Indication)
 - i. Define thresholds:
 1. Low Level $\rightarrow$ Weight < 2 kg
 2. Medium Level $\rightarrow 2$ kg – 5 kg
 3. Full Level $\rightarrow > 5$ kg
 4. Overweight $\rightarrow > 7$ kg
 - ii. LED behavior:
 1. Low $\rightarrow$ Green ON
 2. Medium $\rightarrow$ Blue ON
 3. Full $\rightarrow$ Red ON
 4. Overweight $\rightarrow$ Red BLINKING
- f. Send weight data to Blynk

6. Simulation & Testing in Wokwi / VS Code